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Mechanical pretreatments of different agri-based feedstock in full-scale biogas plants under real operational conditions

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ABSTRACT

Mechanical pretreatments are commonly used devices at full-scale to reduce the size of feedstock and to increase methane production, improving mixing, heat, and mass transfer into the anaerobic digesters of biogas plants. Nevertheless, physical pretreatments require considerable energy. This work aimed to investigate how the energy consumption in different mechanical pretreatments that effectively run under real operational conditions at full-scale level affects physical modifications and anaerobic degradability of agri-based feedstock. Four mechanical pretreatment were investigated: knife milling, hammer milling, extrusion, shredding + hydrodynamic cavitation. After pretreatments, the specific methane yield increased in the range from 1% to 13% and the maximum methane production rate increased in the range from 4% to 48%. Each pretreatment leads to a positive energy balance (0.14–27 kJ kJ^{-1} used). The increasing of energy consumption indicates a more intense pretreatment with an incremental reduction of particles with diameter bigger than 5 mm. On the basis of the calculated regression analysis, a specific combination of additional specific surface area and additional energy in the day of maximum methane production should be expected with mechanical pretreatment that run effectively at full-scale level.

1. Introduction

In Europe, about 74% of the primary biogas energy output in 2017 was produced in biogas plants using agricultural residues, manure, and energy crops, providing significant improvements to the environmental sustainability of the agricultural system by the mitigation of greenhouse gases (GHG) emissions and the production digestate that can be used as valuable natural fertilizer on soil [1].

In the lasts decades, the European biogas sector development was significantly affected by national legislation and regulations capping the use of land for biogas/biomethane production. As a result, the table feeding of agricultural biogas plants are moving from the mainly utilization of crops potentially in contrast with food/feed chain, such as maize, to the valorization of lignocellulose-rich agricultural residues such as manure, straw, corncobs, corn stovers, grass, seeds [2]. Additionally, the interest in crops which are not used directly for human consumption like sorghum and triticale cultivated with sequential cropping system and perennial species grown in marginal and unsuitable for food production soils is increasing. However, while maize silage is highly compatible with the current biogas technology and provides additional advantages (i.e. easy storage and handling, high anaerobic degradability, good mixing in digesters), alternative low-starch content and lignocellulosic feedstocks show technological challenges concerning their processing due to the recalcitrance of lignocellulose [3].

The anaerobic degradability and the biogas production of such substrates mainly depend on their content of cellulose, hemicellulose and lignin. When cellulose and hemicellulose are strictly associated with lignin, it acts as a barrier preventing anaerobic degradability. The structure of lignocellulose is such that cellulose forms a skeleton surrounded by hemicellulose and lignin, functioning as a matrix and

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treats substrates with total solids content typical for energy crops (270–380 g TS kg^{-1}) but processing of substrate with higher total solids content is also feasible. Karuranithy and Muthukumarappan investigated extruder parameters such as screw compression ratio, screw speed, and barrel temperature on grass and switchgrass with moisture contents up to 450 g TS kg^{-1} [12].

Electrical conductivity and sCOD have been proposed for rapid determination of anaerobic degradation [10]. EC in KM treated substrate was 38% higher compared to untreated one while none significantly increasing in EC was observed after HM, EX, SHC pretreatments. The sCOD in KM and HM treated substrate were 90% and 13% higher compared to untreated ones while not significantly increasing in sCOD was observed after EX, and SHC pretreatments. The solubility of agri-based feedstock affects the measurement of both EC and sCOD and subsequently influence the anaerobic degradability prediction meaning that these parameters are not particular suitable as degradability prediction indicators due the heterogeneous nature of agricultural feedstock composition, size and shape.

3.2. Effect of the different mechanical pretreatments on physical characteristics

The particle size distribution of substrates before and after each pretreatment are summarized in Fig. 1. Details about the largest particles fraction, the smallest particles fraction and the overall Specific Surface Area are reported in Table 3.

For each untreated sample, particles larger than 5 mm ranging from 5.5% to 11.5% of the total solids (Table 3). Such kind of solids represent a small proportion of the total amounts of particles, but they mainly affect the rheological properties of the digestate in biogas plants. This is also remarked by Naegele et al. [13] that found in full-scale digesters only 6.7–13.1% of the total particles volume showed a diameter bigger than 4 mm and by Kusch et al. [14] that reported the presence of floating layers in biogas plants fed with large amounts of fibrous material. In practical application, many biogas operators reported the issue of floating layers due to very long particles in the digesters.

The KM untreated seeds represent the substrate with the highest total solids concentration and the lowest specific surface area (SSA) with 3.35 ± 0.37 m^2 $kgTS^{-1}$ among untreated samples while the SHC untreated digestate was the substrate with the lowest total solids concentration and the highest SSA with 28.29 ± 0.74 m^2 $kgTS^{-1}$. This observation is in accordance with the amount of fine solids with diameter lower than 0.3 mm that reach the lowest and the highest value of 2.8 ± 0.7%TS and in 61.4 ± 1.7%TS in seeds and digestate among untreated samples, respectively. The HM and EX untreated agri-based feedstock showed intermediate SSA of 19.88 ± 0.65 and 14.48 ± 2.15 m^2 $kgTS^{-1}$, respectively.

Application of mechanical pretreatments had shown in all cases a decreased percentage of coarsest fraction with diameter bigger than 5 mm and an overall increased percentage of the fine fractions with particle size diameter smaller than 0.3 mm, indicating a successful outcome of the process technologies. This led to an increasing in the SSA ranging from 9% to 189% in the treated samples in comparison to the untreated ones.

KM showed the most relevant effect in terms of particles size reduction (Fig. 1). After KM, a 99% reduction of particles with diameter bigger than 5 mm was observed and a remarkable increasing of 189% of SSA from 3.35 ± 0.37 m^2 $kgTS^{-1}$ to 9.68 ± 1.06 m^2 $kgTS^{-1}$ was observed (Table 3).

HM and EX showed a reduction of 33% and 40% of particles with diameter bigger than 5 mm, respectively. The SSA after HM and EX increased from 19.88 ± 0.65 m^2 $kgTS^{-1}$ and 14.48 ± 2.15 m^2 $kgTS^{-1}$ to 23.17 ± 1.06 m^2 $kgTS^{-1}$ (+17%) and 23.57 ± 1.15 m^2 $kgTS^{-1}$ (+63%), respectively. After HM and EX, the agricultural feedstock showed comparable particles size distribution (Fig. 1) and same SSA (Table 3) probably because similar shear forces acted on the materials during the pretreatment (Supplementary Data S1) and comparable energy consumption was used (Table 4).

A reduction of 50% of particles with diameter bigger 5 mm was achieved with SHC (Fig. 1). The SSA increased from 28.29 ± 0.74 m^2 $kgTS^{-1}$ for the untreated samples to 30.76 ± 0.44 m^2 $kgTS^{-1}$ (+9%) for samples after SHC (Table 3).

Particle size reduction depends on many factors related to the mechanical device and its operation in addition to the substrate properties such as humidity and initial particle size [8].

Coarita et al. reported that high content of long particles increases the yield stress and viscosity of digestate affecting the required energy demand for an appropriate mixing of the digesters [7]. A reduction of the particle size of the feedstock, results in a decrease of the viscosity. Mönch-Tegeder et al. [15] reported a difference of up to 52.5% in viscosity between digestate from a biogas plant fed with untreated feedstocks and one with mechanically disintegrated feedstocks.

In this work, the good mixing in anaerobic digesters at full-scale level was reported by biogas plant's operators confirming the importance of particles size reduction of substrates achieved with mechanical pretreatments.

3.3. SEM analysis

Scanning Electron Microscopy (SEM) was used for the qualitative evaluation on the morphological changes of substrates after each mechanical pretreatment. SEM images are reported in Supplementary data S1.

KM acts cutting fragments and dissociating lignocellulosic biomass. Seeds treated with KM show damaged and cracked surface due to the impact and to the shear between knifes and the sieve placed into KM.

After EX, the treated sample shows a jagged surface due to shear and friction between the screw and the walls of the extruder that represent the main mechanical stresses. Similar effects on morphological degradation generated by extrusion were observed in Hidalgo et al. during the pretreatment of barley straw and vine shoots. After extrusion the structure of plant cells barley straw and vine shoots was damaged, observing the appearance of large cracks [16]. Tsapekos et al. observed similar structural variations on plant tissues between the untreated

Table 3
Effects of mechanical pretreatment technologies on physical characteristics. D = diameter of particles (mm); SSA = Specific Surface Area, KM= Knife Milling; HM=Hammer Milling; EX = Extrusion; SHC=Shredding + Hydrodynamic Cavitation.

Pretreatment technology	Sample	D ≥ 5 mm			D < 0.3 mm			SSA		
		%TS			%TS			m^2 $kgTS^{-1}$		
KM	untreated	5.5%	±	2.1%	2.8%	±	0.7%	3.35	±	0.37
	treated	0.02%	±	0.02%	40.6%	±	5.0%	9.68	±	1.06
HM	untreated	11.5%	±	3.5%	45.3%	±	1.9%	19.88	±	0.65
	treated	7.7%	±	3.4%	50.9%	±	3.6%	23.17	±	1.06
EX	untreated	17.2%	±	3.6%	31.7%	±	5.5%	14.48	±	2.15
	treated	10.3%	±	1.6%	51.3%	±	2.1%	23.57	±	1.15
SHC	untreated	8.4%	±	0.3%	61.4%	±	1.7%	28.29	±	0.74
	treated	4.2%	±	0.1%	67.1%	±	1.3%	30.76	±	0.44

Table 4
Energy consumption, energy balance and benefits in relation to energy consumption of each pretreatment technology.

Pretreatment Technology	Energy consumption		Flow rate	Energy Balance	Additional Energy in the day of max CH_4 production	Additional surface area
	kJ kg^{-1}	kJ $kgTS^{-1}$	t h^{-1}	kJ kJ^{-1}	kJ kJ^{-1} day^{-1}	m^2 kJ^{-1}
KM	33.75	37.94	1.6	27.03	9.31	0.17
HM	19.96	53.13	10.0	18.49	2.35	0.06
EX	15.54	43.91	7.2	6.74	9.19	0.21
SHC	28.80	271.21	2.5	0.14	0.04	0.01

meadow grass and the pretreated one using different pretreatment technology based on extrusion [17].

HM works removing fragments and dissociating pieces from the agri-based feedstock mixture. Impact zone and compression stresses after collisions between hammers and the breaker plate are evident on the surface of the HM treated sample. HM caused a general effect in which some structures are broken-down and some seems to be intact. Ämmälä et al. studied the effects of operational parameters and various screen designs on the fine grinding of *Sphagnum moss* with hammer mill. Although the screen design or operating parameters showed not significant effects on the aspect ratio, the surface of *Sphagnum moss* after hammer milling appeared to be irregular rather than the untreated one [18].

The SHC untreated sample is characterized by rigid and highly orderly structure while the SHC treated sample has a different surface morphology that became roughened and pitted with fibre structure more accessible to microbial anaerobic degradation. Cavitation produces the generation, the growth and the subsequent collapse of microbubbles or cavities in the digestate resulting in very high local energy densities. This produces an extreme induced temperature and pressure on the surface of the sample and a huge shear force which acts to break and disintegrate the lignocellulosic structure. Hidalgo et al. reported SEM images of barley straw and vine shoots taken before and after the cavitation. Cavitation led to breakdown of the lignocellulosic structure of the agri-based feedstock and the structure became loose and disordered [16].

3.4. Effect of different mechanical pretreatments on methane formation

The specific methane yield (BMP) and the maximum specific methane production rate (R_{max}) reached during the laboratory test were studied to evaluate the effects of mechanical pretreatments. Specific methane yield (BMP) and R_{max} are reported in Table 2 while the curves of kinetic in methane formation are detailed in Fig. 2.

The specific methane yield of untreated samples ranging from 123 to 308 Nl CH_4 $kgVS^{-1}$ in accordance with type and origin of substrates used in the biogas plants.

After mechanical pretreatment, an increasing in the specific methane yield was observed in the range from 1% to 13% in the treated samples in comparison to the untreated ones. The specific methane yield after KM and HM increased from 265 ± 11 $NlCH_4$ $kgVS^{-1}$ to 300 ± 10 (+13%) and from 280 ± 16 $NlCH_4$ $kgVS^{-1}$ to 309 ± 1 (+10%), respectively. Slight differences in specific methane yield were observed after EX and SHC in comparison to untreated conditions probably due to the intrinsic biodegradability of the substrates. The substrate is a digestate already processed by anaerobic digestion, therefore a negligible improvement in methane yield after SHC pretreatment was expected.

The increasing in the specific methane yield measured in this study is similar to those observed in other works carried out under optimized conditions at different scale level on agricultural feedstock. Mshandete et al. reported the grinding of sun-dried hemp fibres to a size of 2 mm resulted in a 22% increasing in specific methane production compared to untreated fibres [19]. At lab-scale, Elalami et al. observed knife milling and ball milling of macroalgal residue *(Gelidium sesquipedale)* gave between 10% and 25% higher methane than the raw material [20]. Monch-Tegeder et al. obtained an increasing from 5 to 10% of specific methane production in horse manure treated with a cross-flow grinder at full-scale but under optimized conditions [21]. The application of shearing forces with a rotated plastic sweeping brush against a steel roller significantly increased meadow grass methane yield by 20% under relatively soft operation conditions while a more intense operation was not associated with higher methane yield enhancement [17]. Menardo et al. studied at full-scale level the extrusion effectiveness on rice straw under controlled operational conditions. When extrusion was employed, organic matter degradation was promoted and methane production was significantly raised until 16% more in comparison to untreated rice

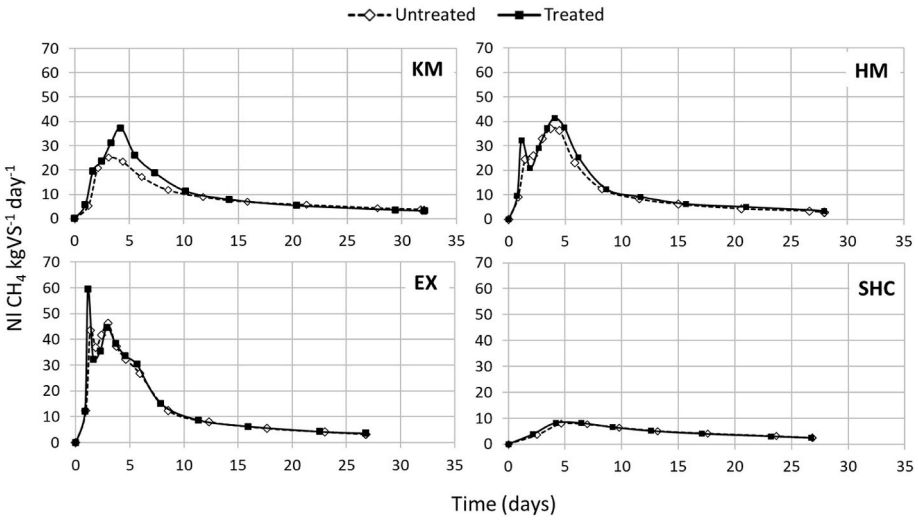

Fig. 2. Methane production rate before and after mechanical pretreatments. Untreated substrate in white dotted lines, treated substrate in black line. KM= Knife Milling; HM=Hammer Milling; EX = Extrusion; SHC=Shredding + Hydrodynamic Cavitation).

straw [22]. Zielinski et al. studied at pilot-scale level that the exposure of substrate prior to anaerobic digestion such as cattle manure and slurry mixed with wheat straw to ultrasound and hydrodynamic cavitation significantly enhanced biogas production by 17–25% in comparison to not-treated substrates offering energy efficiency way for that agricultural residues pretreatment [23]. Ivanchenko et al. found that with increasing intensity of cavitation process occurring in a hydrodynamic disperser (centrifuge) the particles size of a mixture of vegetable waste thickened sewage activated sludge decreases from 50 to 16 μm which causes an increase in biogas yield by ~30% [24]. Nevertheless, none or slight differences in specific methane production after mechanical pretreatment could be observed. Chandra et al. described the grinding of wheat straw and rice straw from 0.75 mm size to less than 0.3 mm lead to a negative results [25] and Coarita et al. reported no significant differences between the BMP values of untreated raw cattle manure in comparison with the substrate treated with hammer mill [26].

On the other hand, kinetic parameters must be considered for the evaluation of mechanical pretreatment on methane formation. After the pretreatments, the maximum methane production rate (R_{max}) increased in the range from 4% to 48% in the treated samples in comparison to the untreated ones. The specific methane production rate after KM and EX increased from 25.13 ± 1.59 $NlCH_4$ $kgVS^{-1}$ day^{-1} to 37.15 ± 2.89 (+48%) and from 46.82 ± 2.00 $NlCH_4$ $kgVS^{-1}$ day^{-1} to 59.55 ± 3.50 (+27%), respectively. This is in accordance with Herrmann et al. that described an enhancement of approximately 18% on the R_{max} with a chopping pretreatment of lignocellulosic feedstocks at farm-scale [27].

In this study, the relative increasing in maximum methane production rate (R_{max}) after pretreatment was related to the relative increasing in the Specific Surface Area (SSA), as shown in Fig. 3.

The relative increasing in the R_{Max} is strictly related to the increasing of the relative SSA, apparently independent by pretreatment technology used and substrates composition. Changes in the surface area is considered the most significant link between comminution and improvements of the biological degradation [5]. Mechanical pretreatments have an effect on particle size reduction improving the biological process kinetics due to the more bio-accessibility of organic matter, as confirmed in other studies [4,16,26,28,29]. To demonstrate the significance of the SSA in degradation processes, a kinetic model for anaerobic degradation of organic substrates has been developed by Palmowski [5, 30]. This model gives a central role to the surface area by setting the degradation rate for substrate hydrolysis proportional to the SSA. The studies concluded that any pretreatment able to increase the SSA is therefore improving the hydrolysis and the entire anaerobic degradation process. The transposition of these results on full-scale biogas plant enables a process with faster methane formation under shorter hydraulic retention time or smaller reactor volumes.

3.5. Relation between analytical parameters and energy consumption

The results about full-scale pretreatment technologies energy consumption and energy balance are listed in Table 4. The energy consumption for the four mechanical pretreatment investigated in this study ranged from 15.5 to 33.8 kJ kg^{-1}. These values are in agree with other works in literature. KM energy consumption of 33.8 kJ kg^{-1} was in the range of 3.6–72 kJ kg^{-1} observed by other studies to treat dried substrates such as corn stover and wheat straw with knife mill [8]. The energy consumption for hammer mill observed in this study was 20 kJ kg^{-1} and it was lower than those of full-scale hammer mills recorded by Coarita et al. that reached 36 kJ kg^{-1} [7] even if the energy consumption is similar when expressed on the total solids content (53 kJ $kgTS^{-1}$ in this work vs 43 kJ $kgTS^{-1}$ in Coarita et al.). The energy consumption of EX was 15.5 kJ kg^{-1}, similar to those reported in Hjort et al. where the extrusion of similar agri-based feedstock consumed 14–36 kJ kg^{-1} of fresh substrate [31]. Garuti et al. investigated the effects of hydrodynamic cavitation of digestate at full scale over the range of about 30–60 kJ kg^{-1} [11] confirming comparable energy consumption also of the present work (28.8 kJ kg^{-1}).

The energy consumption of mechanical pretreatment depends on the feedstock characteristics, the desirable particle size reduction, and the technology used [28]. In this study, the total solids content of untreated substrates ranged from 106.2 to 889.5 gTS kg^{-1} (Table 2), the particles with diameter bigger than 5 mm of untreated substrates was 5.5–11.5% of the total solids (Table 3), and the flow rate of mechanical pretreatment was in the interval from 1.6 to 10 t h^{-1} (Table 4). This work reported four different full-scale pretreatments in which the physical disintegration in relation to the energy consumption was considered suitable by the biogas plant's operators to reach satisfactory process conditions. For KM, energy consumption close to 34 kJ kg^{-1} was used for the abatement of totality of bigger particles. In other cases, such huge reduction of big particles seems not necessary for biogas plants operators to justify much higher energy use.

In this work, the energy consumption of mechanical pretreatment was also related to the reduction of the largest particles with diameter bigger than 5 mm (Fig. 4). Greater the energy consumption, greater the reduction of particles with diameter bigger than 5 mm (Fig. 4). The increasing of energy consumption indicates a more intense pretreatment with an incremental reduction of the quantity of biggest particles. This aspect is relevant to have a positive impact on the rheological properties of digestate because long particle size of substrates affects negatively the yield stress and viscosity of digestate [7]. Additionally, the recommended particle size of lignocellulosic materials should be less than 3 mm for effective microbial accessibility and hydrolysis [32].

Each pretreatment leads to a positive energy balance. The energy balance of the pretreatment with KM, HM, and EX indicates a net income in the range of 7–27 kJ kJ^{-1} used while the SHC energy balance could be considered negligible (0.14 kJ kJ^{-1} used) even though positive (Table 4). These results could be explained with the fact that the best

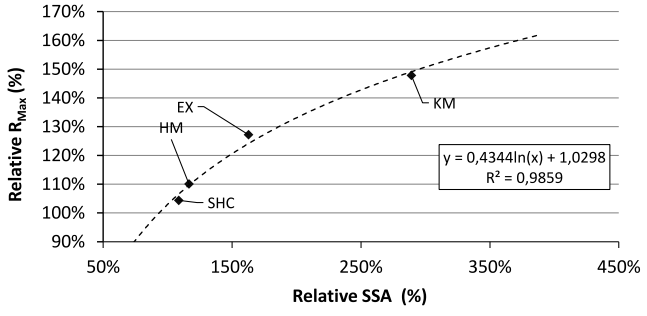

Fig. 3. Relation with the relative increasing of maximum methane production rate (R_{Max}) with the increasing of the relative Specific Surface Area (SSA) among pretreatment technologies.

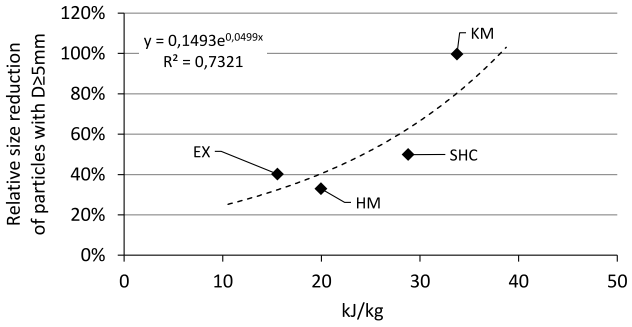

Fig. 4. Relation between energy consumption and 5 mm particle size reduction (D = diameter of particles).

energy balances were obtained with no pre-digested substrates. In particular, the utilization of KM, HM, or EX on lignocellulosic-rich substrates enhanced the availability of new organic substances formerly inaccessible for anaerobic microbial degradation. It is reasonable suppose that to obtain more energy from substrate already digested (i.e.: digestate) the energy consumption with SHC should be more intense but in that case the pretreatment does not generate significant economic benefits.

The additional energy obtained during the day of the maximum methane production was used as indicator of the energetic effectiveness of mechanical pretreatment on the kinetic of the process (Table 4). The best results were obtained with KM and EX with a similar gain of 9 kJ kJ^{-1} used in the day of maximum methane production. HM and SHC led to slight benefits reaching 2.4 and 0.04 kJ kJ^{-1} day^{-1}, respectively.

The additional specific surface area obtained after pretreatment was calculated as indicator of the energetic effectiveness of mechanical pretreatment on the physical disintegration of substrates (Table 4). Among pretreatments, the best results were obtained with KM and EX with a gain ranging 0.17–0.21 m^2 kJ^{-1} used. HM and SHC led to slight additional specific surface area reaching 0.06 and 0.01 m^2 kJ^{-1}, respectively. This is in accordance with the behaviour of the pretreatments shown in Fig. 3 in which the increasing in specific surface area is associated with the increasing of maximum methane production rate. These results could suggest the increasing of specific surface area after pretreatment affected the methane production rate leading to a more solubilization, and bio-accessibility of organic matter as confirmed by other studies [21,26].

Mathematical models were developed to estimate and to predict the energy demand of mechanical pretreatment of lignocellulosic biomass [28]. These studies were limited to the investigation on a single substrate with specific mechanical pretreatment device used in controlled condition (lab/pilot-scale level) neither to consider the methane formation rate effect but only the particles size reduction. Conversely, Coarita et al. proposed the analytical methods to evaluate mechanical pretreatment at full-scale level should be based to both biochemical parameters and physical parameters [7]. The main novelty of the present work is the connection of such biochemical and physical parameters with the energy consumption of four different mechanical pretreatments that effectively run under real operational conditions.

Keeping the same energy consumption among mechanical pretreatment technologies, those with higher additional specific surface area were related to a greater gain in additional energy during the day of maximum methane production, as showed in Fig. 5. On the basis of the reported regression analysis, an indication of a specific combination of additional specific surface area (physical parameter) and additional energy in the day of maximum production (biochemical parameters) should be expected with mechanical pretreatment that run effectively at full-scale level. The combination of aforementioned parameters should be placed upper the trendline when the mechanical pretreatment technology runs effectively at full-scale level. HM is placed below the trendline and a higher additional energy in the day of maximum methane production should be expected in correspondence of the reached additional surface area. That's probably means the energy consumption for HM indicates an intensive particle size reduction even if the benefit on additional energy in the day of maximum methane production rate is lower than those expected. Probably microbiological hydrolysis was not the major rate-limiting step in the substrate treated with HM. A minor energy consumption with a less particle size reduction could be used to try a further optimization of HM at full-scale level.

4. Conclusions

Mechanical pretreatments are commonly used devices at full-scale to reduce the size of feedstock and to increase methane production, improving mixing, heat, and mass transfer into the anaerobic digesters of biogas plants. Nevertheless, physical pretreatments require considerable energy.

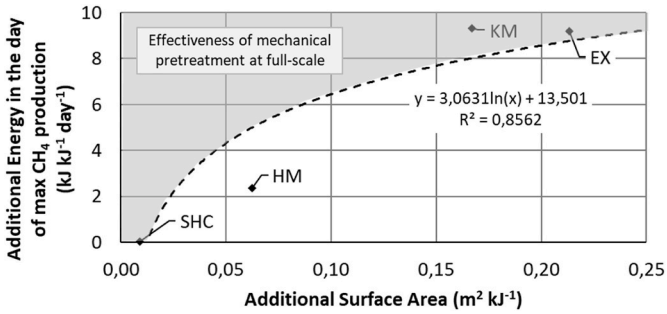

Fig. 5. Relation between additional specific surface area (physical parameter) and additional energy in the day of maximum CH_4 production (biochemical parameter). Mechanical pretreatments placed in the grey area (upper the trendline) indicates effectiveness of mechanical pretreatments at full-scale level.

This work aimed investigate how the energy consumption in four different mechanical pretreatments (knife milling, hammer milling, extrusion, shredding + hydrodynamic cavitation) that effectively run under real operational conditions in four different full-scale biogas plants affects physical modifications and anaerobic degradability of agri-based feedstock.

Application of mechanical pretreatments showed an increasing in the specific surface area (SSA) of substrates ranging from 9% to 189% and an increasing in the specific methane yield (BMP) in the range from 1% to 13%. The relative increasing in the maximum specific methane production rate was strictly related to the increase of the relative SSA, apparently independent by pretreatment technology used and substrates composition.

The specific surface area and the anaerobic degradability of pretreated substrates were related to the energy consumption of each pretreatment technology. On the basis of the calculated regression analysis, a specific combination of additional specific surface area and additional energy in the day of maximum CH_4 production could be expected using mechanical pretreatment with a certain energy consumption. The evaluation of such physical and biological parameters in relation to the energy consumption is a promising approach to identify operational conditions for an effective utilization of knife milling, hammer milling, extrusion, and cavitation in biogas plants faster than the classical and time consuming "trials-and-error" approach used by biogas plant's operators till now.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biombioe.2022.106352.

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